

AI Resume Analyzer + Job Matcher - Complete Interview Preparation Guide

1. Python Fundamentals

- Data Types, Functions, OOP, Exception Handling, File Handling, Modules, Virtual Environments

2. Data Structures & Algorithms

- Arrays, Strings, Linked Lists, Stacks, Queues, Hash Maps, Trees, Time Complexity, Space Complexity

3. Database Management Systems

- SQL, Joins, Indexing, Normalization, Transactions, ACID Properties, Primary/Foreign Keys

4. FastAPI

- REST APIs, GET/POST/PUT/DELETE, Routing, Request/Response Models, Pydantic, Dependency Injection, Middleware

5. Backend Architecture

- Client-Server Architecture, API Design, Authentication, Environment Variables, Configuration Management

6. NLP Fundamentals

- Tokenization, Stop Words, Lemmatization, Stemming, Named Entity Recognition, Text Cleaning, Information Extraction

7. Resume Parsing

- PDF Processing, DOCX Processing, Text Extraction Pipelines, Structured Data Extraction

8. Large Language Models (LLMs)

- Gemini, Prompt Engineering, Context Windows, Tokens, Temperature, Hallucinations

9. Embeddings

- Vector Embeddings, Dense Vectors, Semantic Representation, Embedding Models

10. Semantic Search

- Similarity Search, Embedding Retrieval, Ranking Systems

11. Cosine Similarity

- Vector Similarity, Matching Algorithms, Recommendation Systems

12. ATS Analysis

- Applicant Tracking Systems, Keyword Matching, Resume Scoring, Readability Analysis

13. Machine Learning Basics

- Features, Labels, Training vs Inference, Evaluation Metrics, Classification vs Regression

14. Recommendation Systems

- Resume-Job Matching, Ranking Candidates, Similarity-Based Recommendations

15. RAG (Retrieval Augmented Generation)

- Retrieval, Context Injection, Knowledge Grounding, Vector Search

16. Recruiter Analytics

- Leaderboards, Skill Heatmaps, Candidate Ranking, Data Aggregation

17. React Fundamentals

- Components, Props, State, Hooks, Routing, API Integration

18. Frontend Development

- Vite, TailwindCSS, Responsive Design, Dashboards, Forms

19. Authentication & Security

- JWT Basics, Password Hashing, API Security, CORS, Secrets Management

20. Docker

- Images, Containers, Dockerfile, Docker Compose, Port Mapping, Volumes

21. Deployment

- GitHub, Render, Vercel, CI/CD Basics, Production Environment Variables

22. PostgreSQL

- Schemas, Queries, Indexing, Performance Optimization

23. Software Engineering

- Project Structure, Modular Design, Scalability, Maintainability, Testing

24. System Design

- Frontend → Backend → AI Services → Database Flow, Scaling, Load Handling

25. Interview Questions to Master

- Explain project architecture
- How resume parsing works
- What are embeddings?
- Why cosine similarity?
- How ATS scoring works?
- How RAG works?
- Why FastAPI?
- Why PostgreSQL?
- Why Docker?
- How would you scale the system?